

Intro and background information

Reviewing the available sensors of the android device I felt that for many interesting scientific hypothesis a smart phone does not collect sufficient data, mainly since it is not carried 24/7.

Smart and sports watches however provide far more consistent longitudinal health data.

Therefore I used a Garmin Forerunner 965 as a device and was able to download personal json files to simulate the analytical part.

Early Warning System for deterioration/relaps of Depression

Scientific Question & Hypothesis

Scientific Question: *Can a patient-specific "depressive signature," derived by comparing a historical clinically diagnosed depressive episode against a healthy baseline period, accurately predict the onset of a new depressive deterioration?*

To increase likelihood of success, this design uses a **intra-individual comparison** instead of a population based approach for this highly heterogeneous disease.

By using a known, historical depressive episode as a training anchor, the system learns the patient's specific behavioral degradation pattern to catch a relapse before a clinical collapse occurs.

Hypothesis: An upcoming depressive relapse can be detected earliest by monitoring when a patient's rolling 14-day data approaches the specific Z-score thresholds calculated from their own past depressive episode, rather than relying on generalized population baselines.

The following data based on available sensors are used:

1. **Psychomotor Retardation:** Physical activity (daily step counts) reduces
2. **Decreased Efficiency:** The time required to complete a standardized, recurring route (e.g., a daily commute or a regular running loop, do groceries) increases significantly without a corresponding drop in cardiac load (indicating higher subjective exhaustion)
3. **Circadian Dysregulation:** Sleep patterns fragment (increased night-waking phases, inverted deep-to-REM sleep ratios).

Ideally the data would be substantiated with longitudinal data from Depression questionnaires such as BDI (Beck Depression Inventory) or PHQ-9 (Patient Health Questionnaire-9). However for this exercise these data are not available since these are not sensor related.

2. Sensor Selection & Exact Data File Architecture

To capture the vegetative and psychomotor symptom domains of depression, the system maps two distinct hardware sensors from a wearable tracking device to specific files:

Sensor 1: 3-Axis Accelerometer (Activity Tracking)

- **Target File Name:** `user_wellness_files.json`
- **Theoretical Data Structure & Content:** This file contains an array of daily health summaries. Each object represents a 24-hour epoch.
- **Key Fields for Analysis:**
 - `calendarDate`: The string date anchor used for time-series merging (`YYYY-MM-DD`).
 - `steps`: An integer value counting total daily gait cycles.
 - `activeKilocalories`: A float metric representing metabolic energy expenditure above the resting baseline rate.
- **Clinical Interpretation:** Used to track **Psychomotor Retardation**. During the onset of a depressive episode, a valid signal is a sustained drop in daily steps and active energy expenditure as lethargy increases.

Sensor 2: Optical Photoplethysmography / PPG (Sleep Architecture)

- **Target File Name:** `*_sleepData.json`
- **Theoretical Data Structure & Content:** This file records time-series sleep tracking payloads. Because raw files often record raw timestamp strings rather than pre-computed intervals, the engine must ingest raw sleep boundaries.
- **Key Fields for Analysis:**
 - `calendarDate`: Synchronized date string matching the wake-up morning.
 - `sleepStartTimestampGMT`: ISO-8601 string (`YYYY-MM-DDTHH:MM:SS`) marking sleep onset.
 - `sleepEndTimestampGMT`: ISO-8601 string marking wake time.
- **Clinical Interpretation:** Used to track **Circadian Dysregulation**. Depressive deterioration typically manifests as sleep architecture fragmentation or hypersomnia. By computing the delta between these timestamps, the system captures shifts in total sleep time.

Slide 3: The Analytics Component (Python Architecture)

The Python script needs to do two things:

1. **Phase 1:** Look at the historical healthy vs. depressed data to calculate the personalized threshold.
2. **Phase 2:** Run a rolling check on new incoming data to see if it crosses that specific threshold.

The Mathematical Anomaly Detection Principle

- The algorithm establishes a healthy baseline using a known stable period (e.g. April–June 2026) to find the mean and standard deviation. It then measures how far the historical depressive episode (e.g. Oct 2024–March 2025) deviated from that norm by calculating an intra-individual Z-score:
- These historical Z-scores become the personalized thresholds. If new incoming 14-day rolling data moves **75% of the way** toward that past depressive signature, a warning flags.

Example Results with alert status:

```
*** === THEORETICAL ANALYTICS RESULTS ===
Calculated Steps Warning Threshold (Z-Score): -4.54
Calculated Sleep Duration Warning Threshold (Z-Score): +2.65

=== PREDICTIVE INFERENCE IN ACTION ===
ALERT STATUS: STABLE.
```